

Pilot-Aided Channel Estimation Scheme for NR-V2X Speed Emulation Technique

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Technische
Universität Wien

Institute of
Telecommunications



Introduction

- vehicular applications
 - improve transportation quality and safety



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 - improve transportation quality and safety
 - have high communication requirements
 - dynamic environment and high mobility



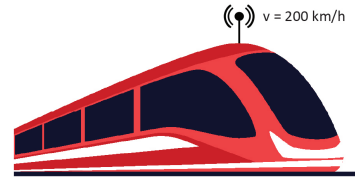
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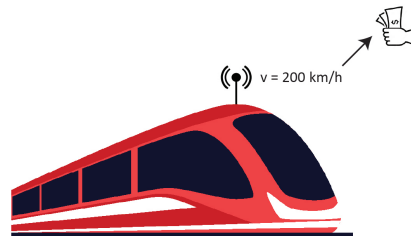
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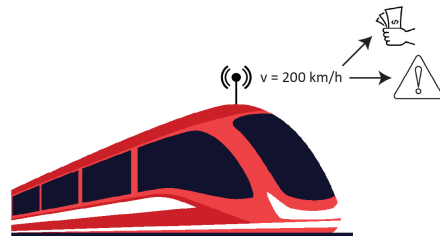
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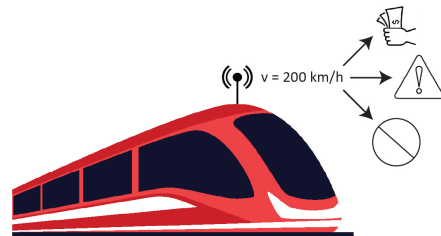
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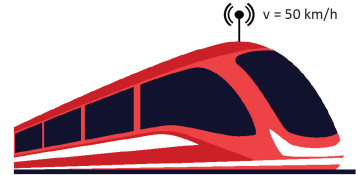
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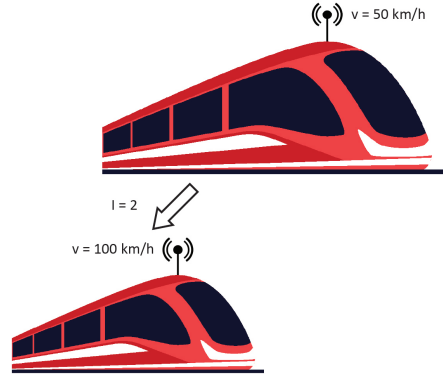
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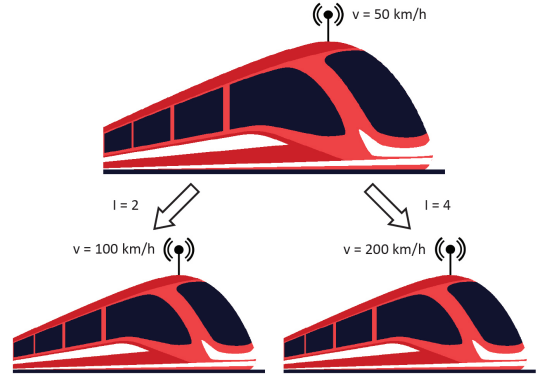
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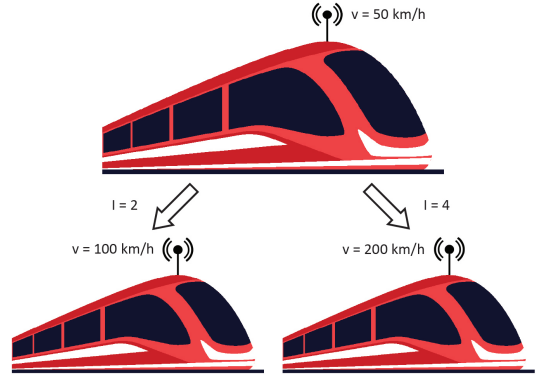
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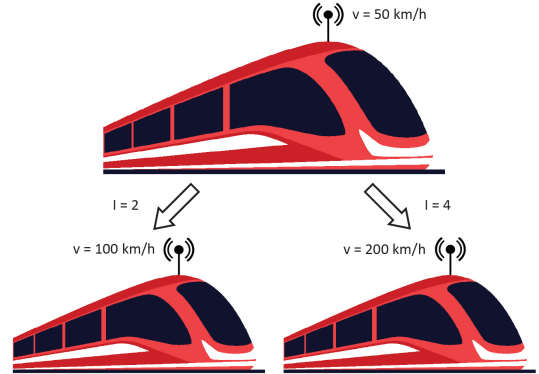
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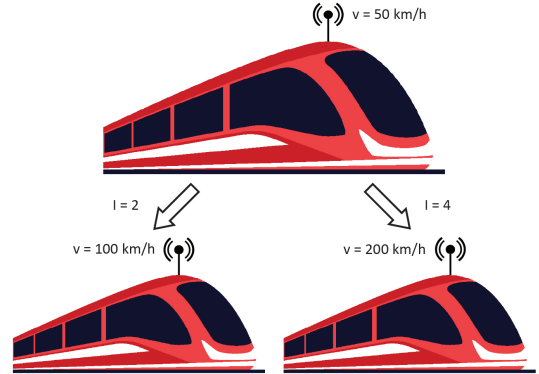
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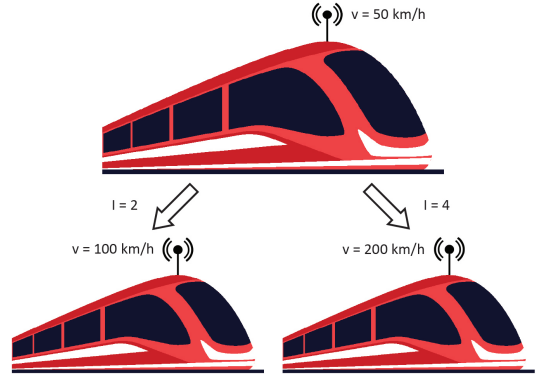
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 - link level simulations and high-speed measurements
 - bit error ratio (BER)

Time-Stretching Technique

Channel Estimation

Validation

Summary and Conclusion

Time-Stretching Technique - how it works ?

- OFDM within time-variant multipath channel

T – symbol duration

I – interpolation factor

f_c – carrier frequency

T^I – adjusted symbol duration

D_n – normalized Doppler spread

c – speed of light

$$D_n = f_{D_{\max}} \cdot T$$

ICI ... Inter-Carrier-Interference

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Time-Stretching Technique - how it works ?

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- rewrite maximum Doppler frequency
- emulate velocity v^I while conducting measurements at actual velocity v

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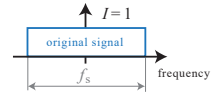
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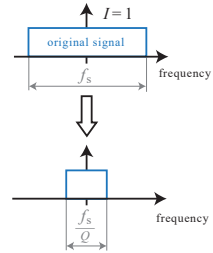
- implementation within OFDM system



OFDM ... Orthogonal Frequency Division Multiplexing

Time-Stretching Technique - Implementation

- implementation within OFDM system
 - decrease the subcarrier spacing

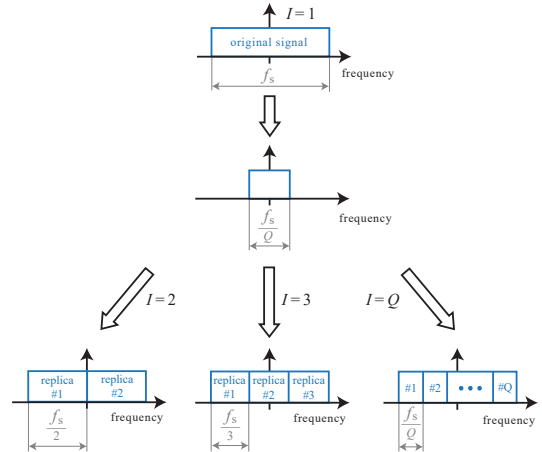


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Time-Stretching Technique - Implementation

- implementation within OFDM system
 - decrease the subcarrier spacing
 - repeat to occupy the same bandwidth

OFDM ... Orthogonal Frequency Division Multiplexing



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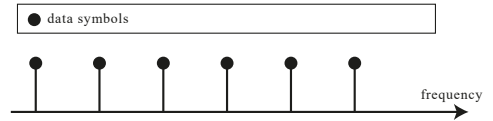
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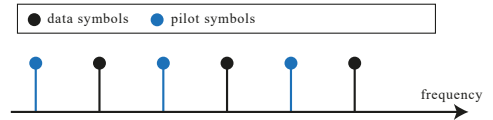
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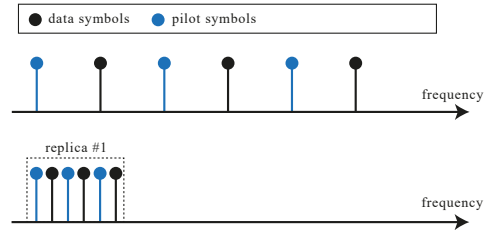
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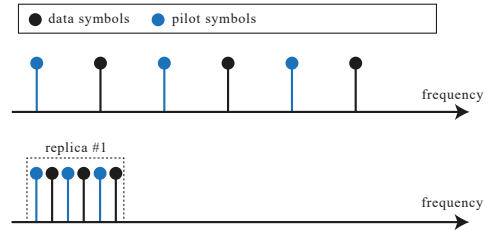
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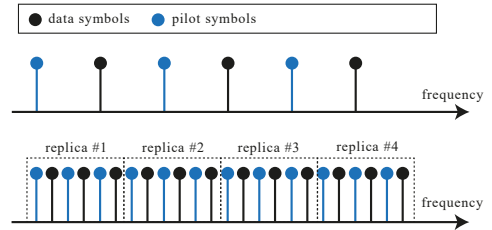
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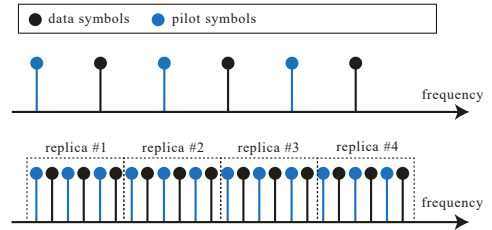
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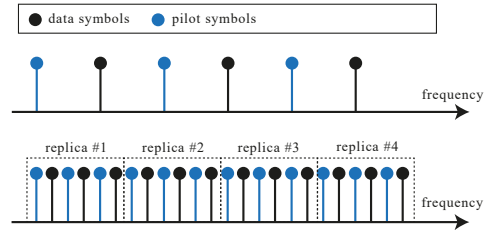
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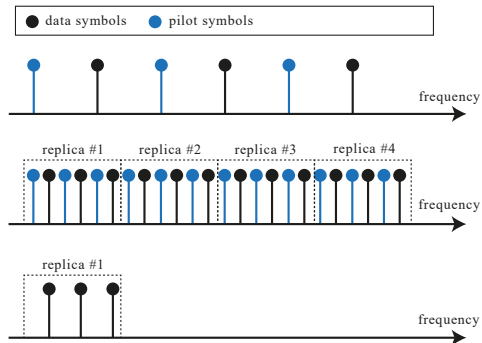
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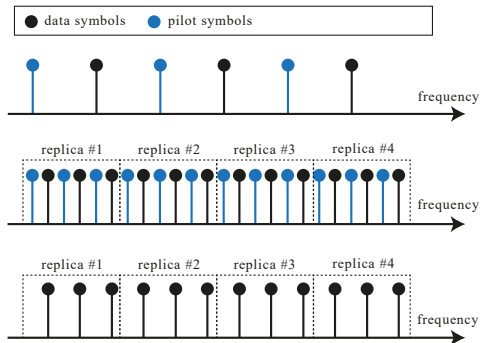
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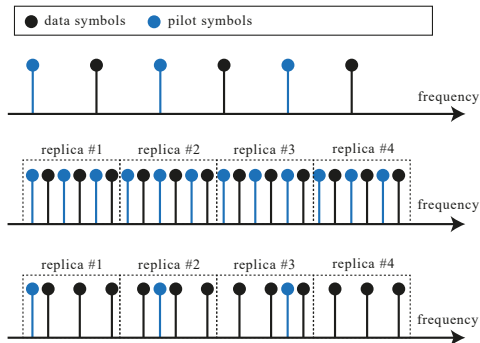
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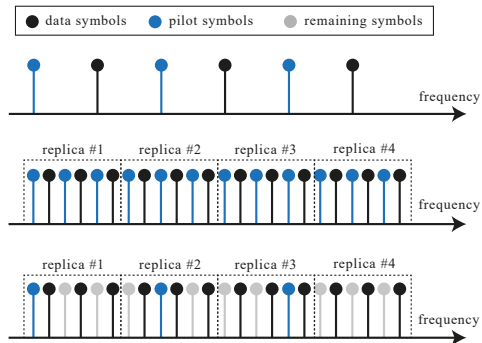
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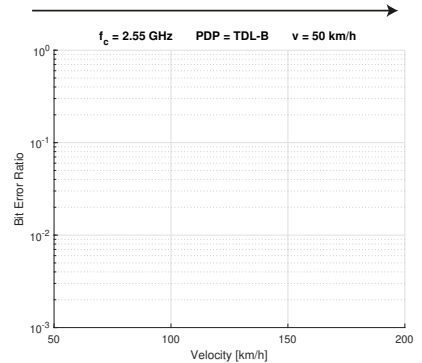
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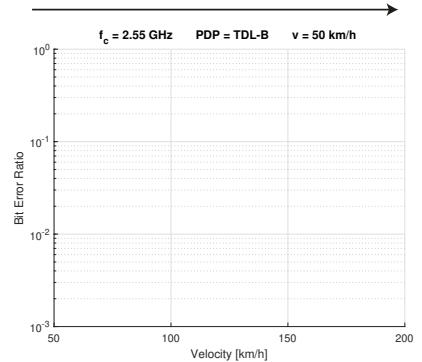
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BER ... Bit Error Ratio
PDP ... Power Delay Profile
TDL ... Tapped Delay Line

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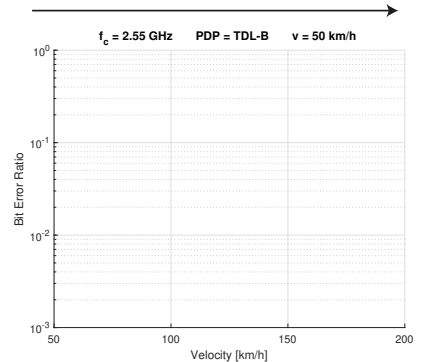
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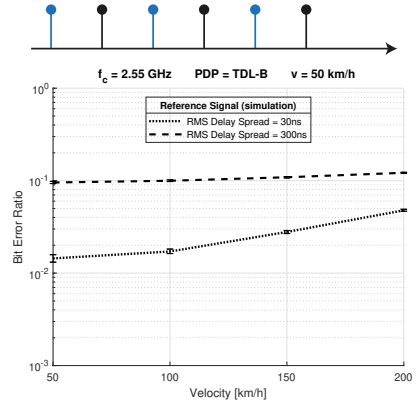
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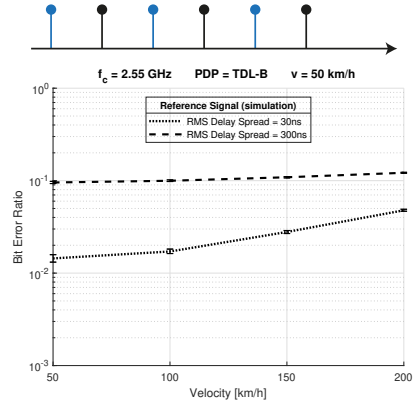
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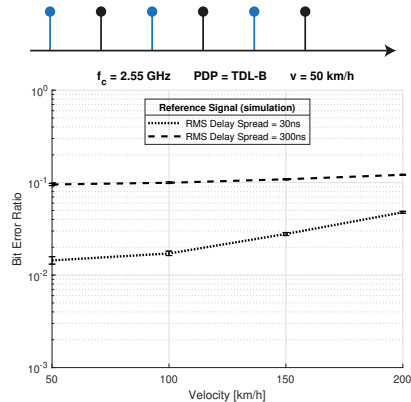
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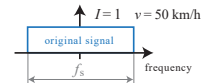
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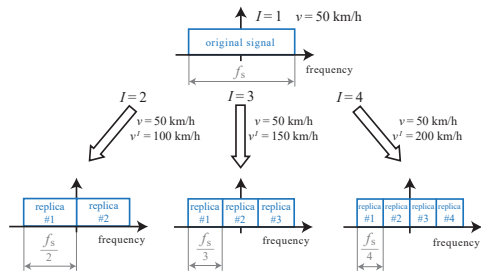
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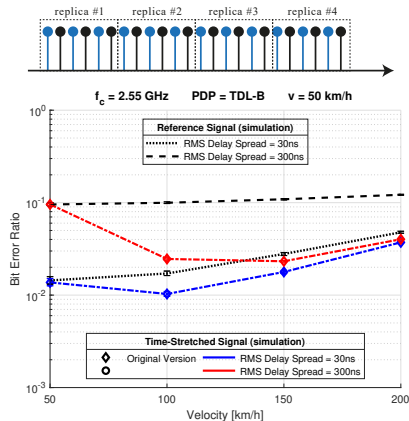
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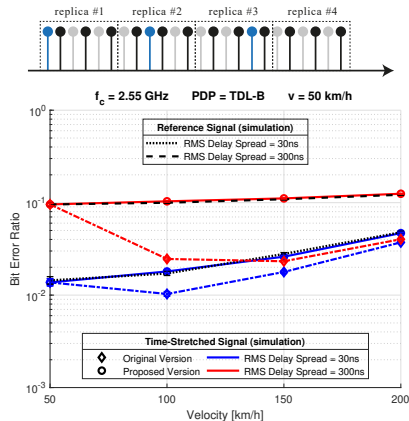
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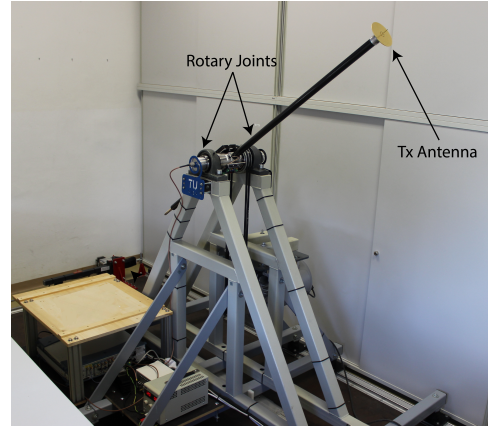
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Measurement-Based Validation - Setup

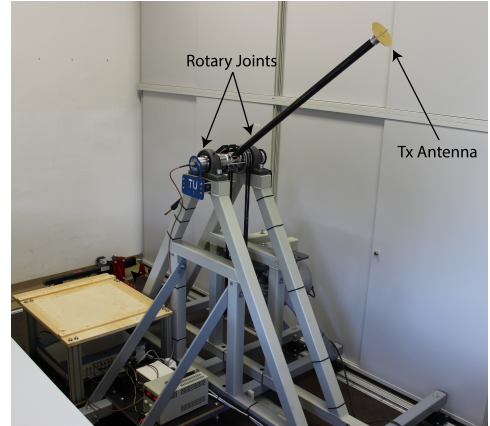
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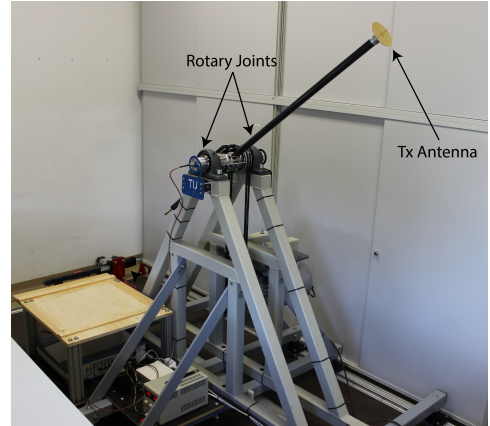
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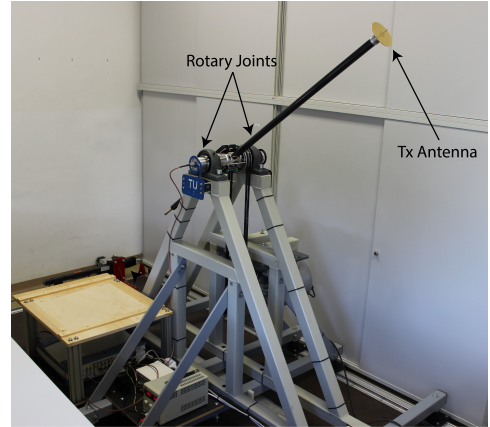
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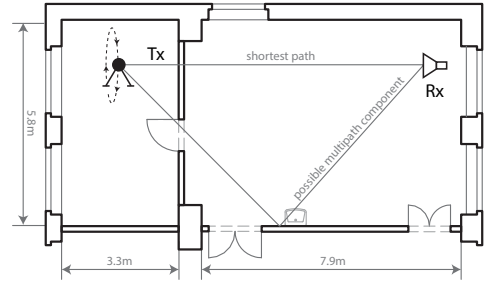
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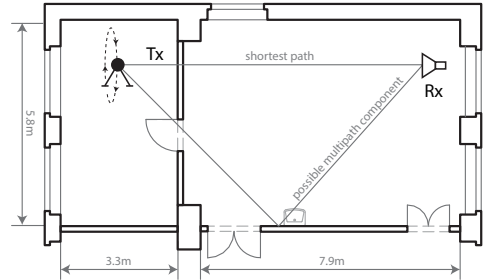
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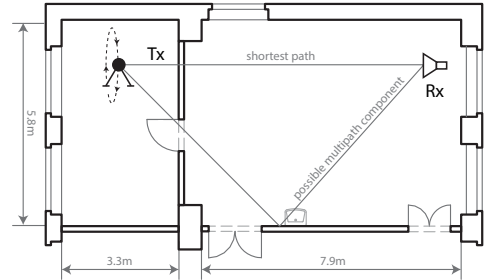
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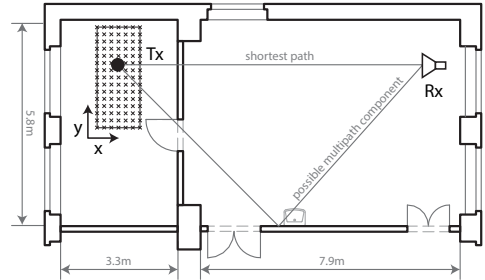
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The 16th European Conference on Antennas and Propagation (EuCAP 2022)

Measurement-Based Validation - Setup

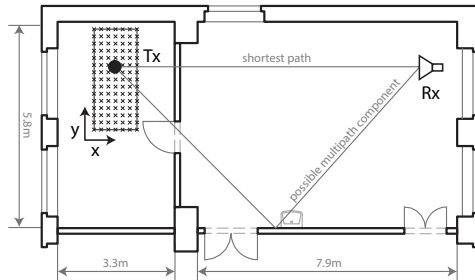
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 - measurement testbed described in [2]
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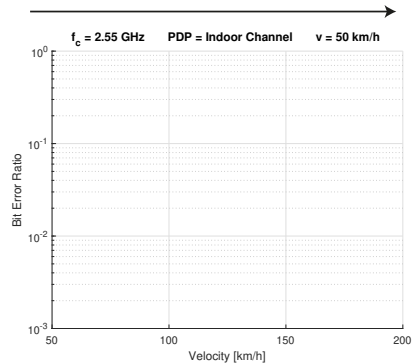
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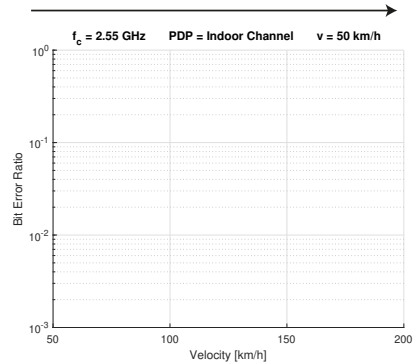
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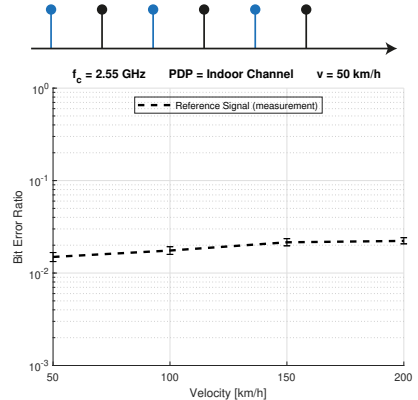
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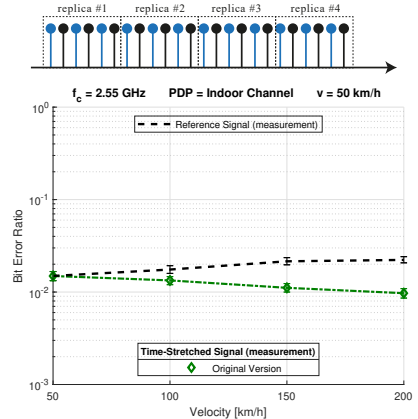
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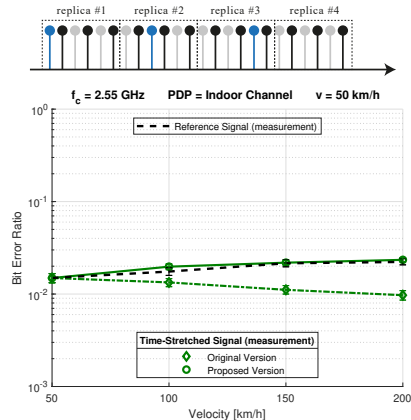
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 - fair comparison to the original system

Pilot-Aided Channel Estimation Scheme for NR-V2X Speed Emulation Technique

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2022 Asilomar Conference on Signals, Systems, and Computers



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